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A deep learning and image enhancement based pipeline for infrared and visible image fusion

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ABSTRACT

It is difficult to use supervised machine-learning methods for infrared (IR) and visible (VIS) image fusion (IVF) because of the shortage of ground-truth target fusion images, and image quality and contrast control are rarely considered in existing IVF methods. In this study, we proposed a simple IVF pipeline that converts the IVF problem into a supervised binary classification problem (sharp vs. blur) and uses image enhancement techniques to improve the image quality in three locations in the pipeline. We took a biological vision consistent assumption that the sharp region contains more useful information than the blurred region. A deep binary classifier based on a convolutional neural network (CNN) was designed to compare the sharpness of the infrared region and visible regions. The output score map of the deep classifier was treated as a weight map in the weighted average fusion rule. The proposed deep binary classifier was trained using natural visible images from the MS COCO dataset, rather than images from the IVF domain (called "cross domain training" here). Specifically, our proposed pipeline contains four stages: (1) enhancing the IR and VIS input images by linear transformation and the High-Dynamic-Range Compression (HDRC) method, respectively; (2) inputting the enhanced IR and VIS images to the trained CNN classifier to obtain the weight map; and (3) using a weight map to obtain the weighted average of the enhanced IR and VIS images; and (4) using single scale Retinex (SSR) to enhance the fused image to obtain the final enhanced fusion image. Extensive experimental results on public IVF datasets demonstrate the superior performance of our proposed approach over other state-of-the-art methods in terms of both subjective visual quality and 11 objective metrics. It was demonstrated that the complementary information between the infrared and visible images can be efficiently extracted using our proposed binary classifier, and the fused image quality is significantly improved. The source code is available at https://github.com/eyob12/Deep_infrared_and_visible_image_fusion.

1. Introduction

Infrared and visible image fusion is an emerging technique for generating an informative fused image that is more reliable and effective, owing to the wide availability of low-cost cameras for infrared (IR) and visible (VIS) image capture. In human and computer vision, by integrating complementary information about the visual content from these two types of source images is necessary. Generally, visible images are visually pleasing and have a high spatial resolution and many edge details. However, visible lighting imaging is susceptible to poor illumination, fog, and poor weather conditions. Infrared imaging capturing thermal images are robust to these disturbances but they typically have low resolution and poor texture. Therefore, fused images with clear visual objects and other clear objects may provide richer information for downstream applications, such as remote sensing, medical diagnosis, surveillance, military, machine vision, and photography, in all day and all weather.

Generally, IR and VIS image fusion methods involve three stages, as shown in Fig. 1 including (1) feature extraction, (2) feature fusion, and (3) fusion image reconstruction. In the simplest case, an infrared image and a visible image can be fused by directly taking the pixel-wise maximum between these two images, where the feature extraction stage can be treated as an identity transformation(i.e., the original pixel intensity is treated as a feature), the feature fusion stage is a maximizing operation, and the fusion image reconstruction stage can be treated as an identity transformation. Different fusion methods have been proposed using different strategies in one of these three stages.

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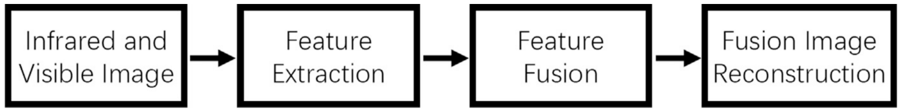

Fig. 1. Three stage fusion process.

In the last decade, many image fusion (IVF) methods are proposed, and they can be classified into two classes in terms of feature extraction [1]: (1) non-feature-learning-based approaches (traditional methods) and (2) feature-learning-based techniques (modern deep learning approaches). Recent IR and VIS image fusion techniques are reviewed in [1–3].

Traditional non-deep feature learning fusion techniques include multiple scale transformation, sparse representation, subspace, and saliency methods [1–3], such as discrete wavelet transformation (DWT) [4], discrete cosine harmonic wavelet transform (DCHWT) [5], gradient transfer and total variation minimization (GTF) [6,7], adaptive sparse representation (ASR) [8,9], low-rank decomposition [9–11], non-subsampled contour transformation (NSCT) [12], filter-based multiscale decomposition [13–15], gradient saliency map [16,17], and nonsubsampled pyramid filters (NSP) [18].

Modern deep learning-based IVF methods use different models, such as convolutional neural network (CNN) [19–23], attention mechanisms [24–27], and auto-encoders(AE) [28–31], generative adversarial networks(GAN) [32–39], or transformer [33,40–43].

In addition, by combining modern deep learning with traditional methods, some researchers have proposed mixed methods, such as CNN plus low rank [21] and CNN plus multiscale decomposition [44].

The traditional fusion methods use hand crafted features in feature extraction stage; however, modern fusion approaches take advantage of the automatic feature learning power of deep convolutional neural networks [1]. Furthermore, compared with traditional methods, deep-network-based methods require considerably less manual design work [1].

However, most modern fusion methods use self-supervised techniques, in which input images are used as pseudo-target images, due to the unavailability of ground truth target fusion images. Therefore, the output of a deep fusion network may deviate from the unavailable ground-truth target fusion image [1]. Furthermore, image quality enhancement is seldom considered in existing fusion methods [1–3], which leads to poor contrast in the fused image as shown in our experimental section. Although, much progress has been achieved in IVF, IVF task is still a challenging problem [1].

To address these problems, we combined modern deep learning with traditional image processing to build a simple and efficient IVF pipeline. We converted the IVF problem into a binary sharpness supervised classification problem between IR image and VIS images. A special deep classifier is specially designed and trained using a cross-domain supervised training strategy, in which the training images are naturally visible images from the MS COCO dataset in the visible domain rather than those from the IVF domain (so-called "cross-domain training" in this study). The trained network accepts the IR and VIS input pairs and outputs the classification score map as a weight map. A weight map is used for the weighted average fusion stage. The main contributions of this study are as follows:

- A simple IVF pipeline was proposed by combining modern deep learning with traditional image processing.
- The IVF problem was converted into a binary supervised classification problem.
- A special multi-path deep fusion neural network was designed.
- A cross-domain training methodology was proposed to train our proposed network in a supervised manner.
- The superior performance of our proposed IVF pipeline was extensively verified using public datasets and providing source codes for research community.

The remainder of this paper is organized as follows. In Section 2, the related work and motivation are presented. The proposed method is described in Section 3. The experimental results and discussion are shown in Section 4. Finally, the conclusions are presented in Section 5.

2. Related work and motivation

Most traditional IVF methods manually design fusion rules [1]. Source images are transformed into other spaces or feature spaces using a linear or nonlinear transformation. In the transformed feature space, the features of source images were manually fused using fixed rules. Then, the fused features are back-transformed to the input space to obtain fused images, such as discrete wavelet transformation [4], discrete cosine harmonic wavelet transform [5], gradient transfer and total variation minimization [6,7], adaptive sparse representation [8,9], low-rank decomposition [9–11], and non-subsampled contour transformation [12], filter-based multiscale decomposition [13–15], gradient saliency map [16,17], and nonsubsampled pyramid filter [18]. sparse representation, low-rank decomposition and multiscale decomposition can also be considered as forms of transformation because they convert images from a spatial space to another representation space.

Although traditional IVF methods are simple, transparent, and interpretable, designing manual fusion rules in complex feature spaces is difficult. For better fusion performance, many deep learning based IVF methods are proposed [19–33,33–43]. Most of these methods use self-supervised information (input images are treated as supervising images) to train deep networks because of a shortage of ground-truth target fusion images. In contrast to traditional IVF methods, deep learning IVF methods can seamlessly implement feature extraction, feature fusion, and fusion image reconstruction in an end-to-end manner to avoid manual design work. However, deep learning can only be used in some parts of IVF problem rather than in all parts [1].

In [19–23], CNN was employed to extract feature, fuse feature, and reconstruct fusion images. To improve the performance of CNN-based IVF methods further, an attention mechanism was embedded into the CNN [24–27]. In AutoEncoder-based IVF methods [28–31], an encoder is used to extract features, and a decoder is used to reconstruct the fusion image. Feature fusion can be implemented using a manually designed rule or an automatic CNN. GAN-based IVF methods [32–39] mostly use a generator to generate a fusion image and a discriminator to distinguish between the source image (treated as a pseudo-target image again) and the fused image. Transformer-based IVF methods have been proposed to utilize long-range information [33,40–43].

Although deep-learning-based IVF methods may use different models, such as CNN, GAN, or transformer, most of them are unsupervised, and their loss function is normally constructed between the fused image and source images by treating the source image as a pseudo-target image. Therefore, it may lead to overfitting to a pseudo-target and deteriorate the performance of unsupervised IVF methods [1].

In addition, some researchers have proposed mixed methods, such as CNN plus low rank [21], and CNN plus multiscale decomposition [44] by combining modern deep learning with traditional methods.

In [45,46], the authors converted the multi-exposure image fusion problem into a blur classification problem, in which the well-focused part was considered sharper than the defocused part. The authors of [45,46] used the output classification score of their deep network as a weight for the weighted averaging fusion rule. Inspired by this, we converted the IVF problem into a blur detection problem and proposed a new simple deep learning and image-enhancement-based IVF pipeline

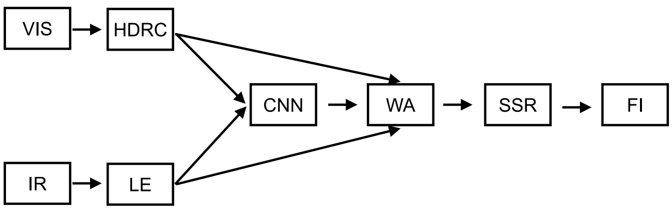

Fig. 2. Proposed Pipeline. VIS: visible input image; HDRC: high dynamic range compression enhancement; IR: infrared input image; LE: linear enhancement; CNN: convolutional neural network; WA: weighted average; SSR: single scale retinex enhancement; FI: fused image.

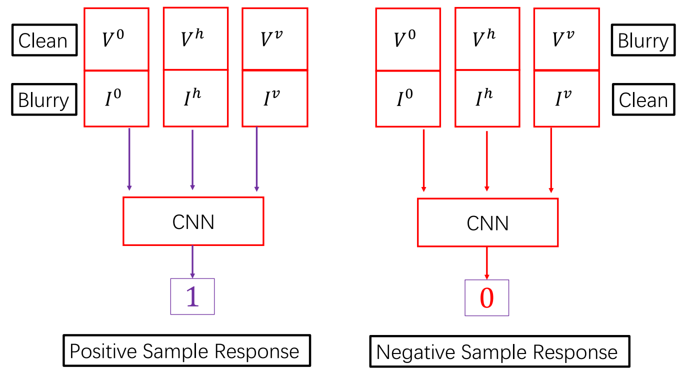

Fig. 3. Illustration of input and output of our proposed CNN with positive sample or negative sample: visible modality (V^0, V^h, V^v) in first row; infrared modality (I^0, I^h, I^v) in the second row.

under the assumption that the informative part may have less blur and that the non-informative part may have more blur in the IVF domain.

The methods for multi-exposure image fusion [45,46] cannot be simply extended to solve the IVF problem because the IVF and multi-exposure fusion problems are essentially different from each other. the multi-exposure fusion problem focuses on the focus problem, whereas the IVF problem focuses on the multi-modality (visible and infrared) image fusion problem. Our work differs from [45,46] in terms of the following:

1. We solved the IVF problem, which is a different challenging problem.
2. We proposed an IVF pipeline that combines modern deep learning with traditional image processing (i.e., image enhancement).
3. Our proposed blur classification network has a residual structure, which is more efficient for training.
4. In IVF, our pipeline achieves better experimental performance than other methods [45,46], which are tailored to the multi-exposure fusion problem.

Different from most of existing IVF methods, our proposed pipeline not only converts the unsupervised IVF problem into a supervised classification problem but also shares the advantages of both the modern deep learning approach and the traditional method (transparent weighted average fusion rule and image enhancement). Furthermore, the good performance of the proposed pipeline in this study verifies the validity of our assumption that a modality with a sharp image provides more salient information than a modality with a blurred image. The method proposed not only fuses salient information from two modalities but also controls the image contrast and quality.

3. Proposed pipeline

Our proposed pipeline, as shown in Fig. 2, contains four stages: (1) enhancing IR and VIS input images using linear transformation and the high-dynamic-range compression (HDRC) method, respectively; (2) inputting enhanced IR and VIS images to the trained CNN to obtain the weight map; (3) using a weight map to obtain the weighted average of enhanced IR and VIS images; and (4) using single scale Retinex (SSR) to enhance the fused image to obtain the final enhanced fusion image. Further details of the proposed pipeline are shown in Fig. 2 and are presented in the following subsections.

3.1. Proposed network

In this study, two patches in the same spatial position from the visible image and the corresponding infrared image are denoted by V^0 and I^0, respectively. V^h is the horizontal derivative of V^0, V^v is the vertical derivative of V^0, I^h is the horizontal derivative of I^0, and I^v is the vertical derivative of I^0. The tuple $t = (V^0, I^0, V^h, I^h, V^v, I^v)$, consisting of six patches, is labeled 1 if V^0 (visible modality) is less blurry than I^0 (infrared modality). Otherwise, it is labeled as 0. The tuple t is the input of our proposed deep CNN. The input and output of our proposed CNN are shown in Fig. 3, where the tuple $t = (V^0, I^0, V^h, I^h, V^v, I^v)$ is organized into three input pairs (visible modality in upper part, infrared modality in lower part) to our proposed CNN with three input paths.

The proposed CNN is a multipath residual convolutional network, as shown in Fig. 4. Our network contains five paths $p^i, i = 1, 2, \dots, 5$ for six patch inputs V^0, I^0, V^h, I^h, V^v, and I^v. Each input patch measured 32 × 32 pixels. V^0 and I^0 were concatenated in row dimensions, as shown in Fig. 3, and fed to the first path consisting of one convolution layer and four residual blocks, as shown in Fig. 4. V^h and I^h concatenated in the row dimension, as shown in Fig. 3, were fed into the second path consisting of one convolution layer and three residual blocks, as shown in Fig. 4. V^v and I^v are also concatenated in row dimensions, as shown in Fig. 3, and fed to the third path, consisting of one convolution layer and three residual blocks, as shown in Fig. 4. Subsequently, the output of the second path and the third path is concatenated in the channel dimension of the fourth path and consists of one concatenation layer and one residual block. The outputs of the first and fourth paths are concatenated in the channel dimensions of the fifth path, which consists of one concatenation layer and one fully connected layer with one node. In the fifth path, the input is mapped to a pure decimal number using a fully connected layer with a sigmoid activation function. The blue numbers beside each layer/block (such as '64 × 32 × 64') indicate the data/image dimension size.

In each path, identical residual blocks and convolutional residual blocks exist as residual blocks, and the structures of which are illustrated in Fig. 4. In an identical residual block, the size of the output dimension was identical to that of the input dimension. In a convolutional residual block, the output dimension is half of the input dimension size. As shown in Fig. 4, an identical residual block contained one path and two convolutional layers. Our convolutional residual block contains two paths and three convolutional layers (see Fig. 5).

As shown in Fig. 3, our proposed binary classification network can predict a clearer patch between the VIS and IR patches using the output score s: $s = 1$ for clearer VIS patches and $s = 0$ for clearer IR patches.

The proposed network has a multipath residual structure. For the input patch pair (V_0, I_0), we concatenate them in the row dimension rather than in the third dimension (channel dimension) in order to reduce the number of potential filters in the network because the channel dimension of the filters increases with an increase in the input channel dimension. We also used different paths for different input patch pairs (gray scale pair (V^0, I^0) for path 1, horizontal derivative pair (V^h, I^h) for path 2, and vertical derivative pair (V^v, I^v) for path 3) because we expected that different paths would focus on different input styles (gray scale intensity, horizontal derivative and vertical derivative). It is also possible to concatenate all patches in the third dimension (channel dimension) to a single input path. However, we choose a multipath network to take advantage of the different paths and adapt to different input styles.

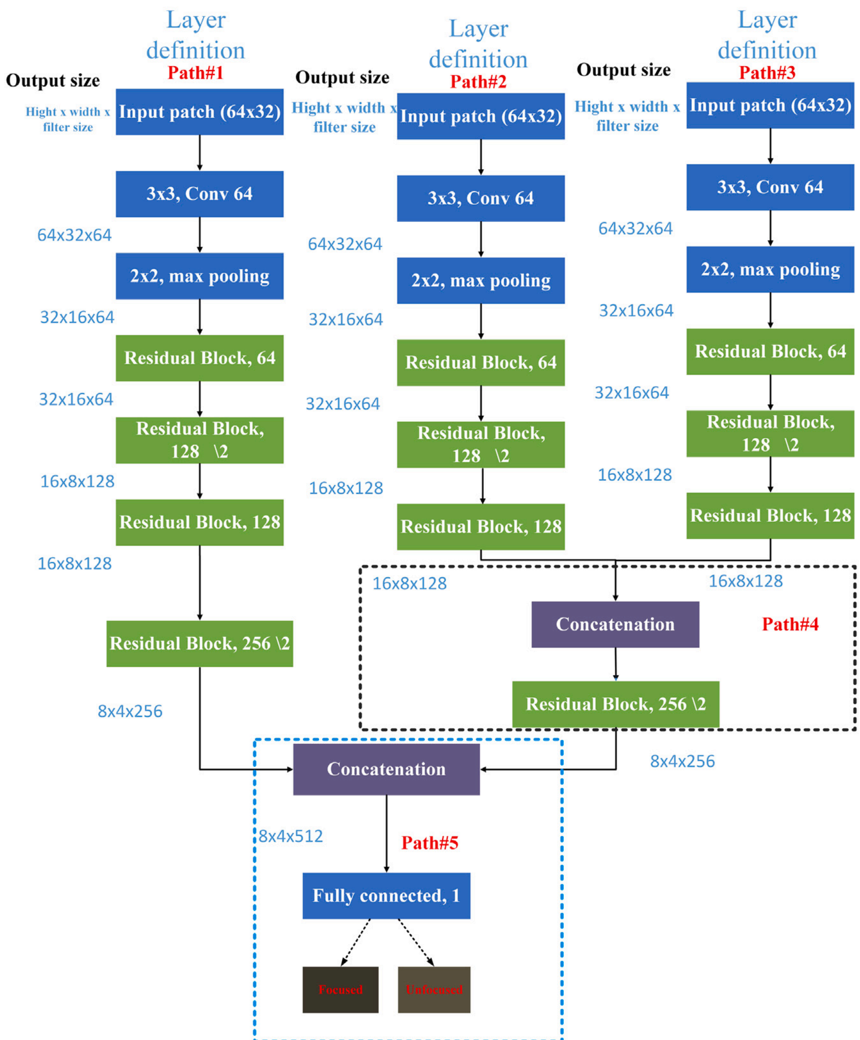

Fig. 4. Proposed multi-path residual network structure.

3.2. Cross domain training strategy

Typically, IVF datasets do not provide clear or blurred labels. We used naturally images from the COCO 2014 dataset and a Gaussian filters of different sizes were used to simulate the blurred-image generation process. The training patch pairs were extracted from the clear images and their corresponding blurred images. We observed that this cross-domain training strategy, which trained in the COCO 2014 dataset and tests in the visible and infrared image fusion datasets, work very well in the IVF problem.

Specifically, we randomly selected 2200 high-quality images from the COCO 2014 dataset and converted them to grayscale images. Then, each grayscale image was blurred by Gaussian filters of four different sizes, $9\times9, 11\times11, 13\times13$, and 15×15 pixels to produce four versions of blurry images. Both horizontal and vertical derivative images of all clear and blurry images were computed. From these images, we randomly extracted 1,000,000 and 300,000 patches for training and validation, respectively. The patch size was 32×32 pixels. A training tuple t comprising six patches is shown in Fig. 6.

To learn the parameter values θ of our binary classification CNN, we used cross entropy loss and solved the following optimization problem:

$$arg_\theta \min L(\theta) = arg_\theta \min \sum_{i=1}^{N} f^{l_i}(\theta, x_i)(1-f(\theta,x_i))^{1-l_i}, \tag{1}$$

where N, x_i, l_i, and $f(*)$ are total number of samples, ith input sample, ith label, and network function, respectively.

3.3. Weight map generation

After the blur detection network mentioned above was trained, it was used to generate a weight map for each pair of visual and infrared images using a sliding window. Specifically, the horizontal and vertical derivative images of the visual and infrared images were computed. The patch tuple t centered around each pixel is the input to the network and the output of the network is the weight of the pixel. Thus, we can obtain the entire weight map for each visual and infrared image pair using a sliding window, as shown in Fig. 7. In Fig. 7, a large weight in the weight map indicates a greater contribution from the visible image and a small weight indicates a greater contribution from the infrared image. The person in the infrared image has more contribution to the fused image since the weights corresponding to the person are small.

3.4. Original image enhancement

Owing to poor lighting conditions at night, we proposed enhancing the visible and infrared images before fusion using different methods. We used HDRC based on a guided filter(GF) [47] to enhance visible images for night vision. The original visible image I_v with width w and

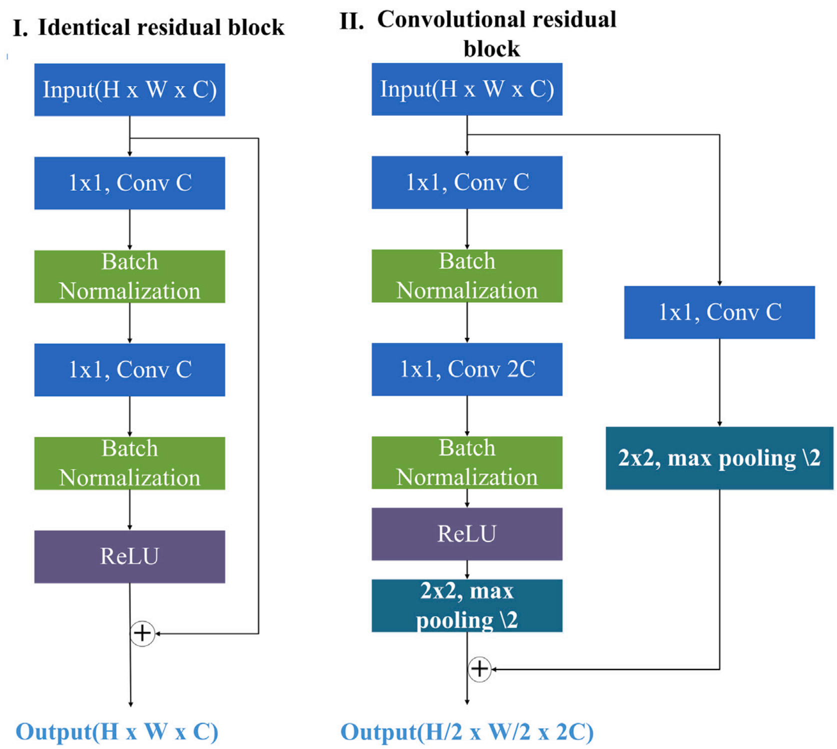

Fig. 5. Structure of our proposed two classes of residual blocks: identical and convolutional residual blocks.

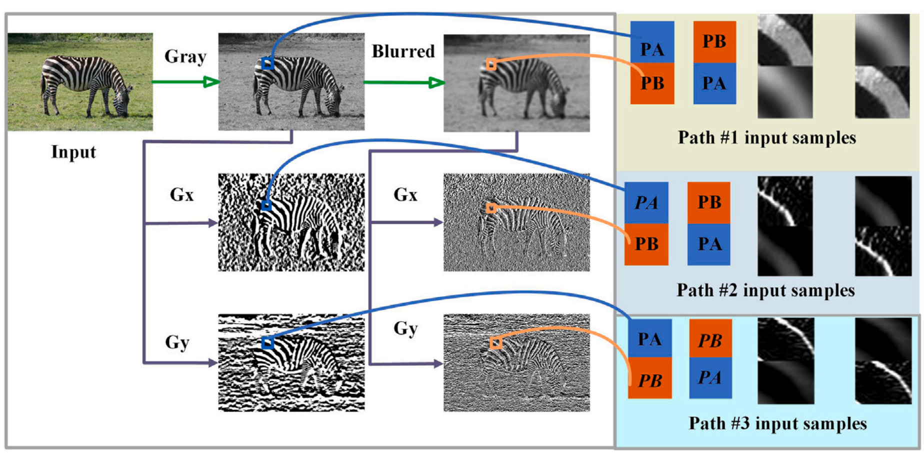

Fig. 6. One training tuple example. Gx: horizontal derivative; Gy: vertical derivative; PA: 32 × 32 patch A; PB: 32 × 32 patch B.

height h was normalized to the range $[0, 255]$. The filtered image I_f is computed as

$$I_f = GF_{r,\epsilon}(I_v), \tag{2}$$

where GF is a guided filter (refer to [47] for more information) with filter size $r = \lfloor 0.4 * \max(w, h) \rfloor$ and edge-preserving parameter $\epsilon = 0.01$.

Following Durand et al. [47], we can obtain the base layer I_b and detail layer I_d as follows:

$$I_b = \log(I_f + \eta) \tag{3}$$

$$I_d = \log(I_v + \eta) - I_b, \tag{4}$$

where log denotes the natural logarithm operator and $\eta = 1$ is used to prevent the log value from being negative.

The dynamic range compression is performed on the base layer I_b with a scaling factor β and the enhanced image I_e is obtained as

$$I_e = \exp(\beta I_b + I_d + \gamma), \tag{5}$$

where $\beta = \frac{\log(T)}{\max(I_b)-\min(I_b)}, \gamma = (1-\beta)\max(I_b)$, and the target base contrast T is set to 4. It can be observed that β is less than 1, and that γ is greater than 0. These settings can considerably enhance the visible light image under poor lighting conditions, as shown in Fig. 8. This figure shows that the enhancing process worked well.

For the enhancement of the infrared image I_i, we used a linear transformation to obtain enhanced one I_i^e as follows:

$$I_i^e = \frac{I_i - \min(I_i)}{\max(I_i) - \min(I_i)} \tag{6}$$

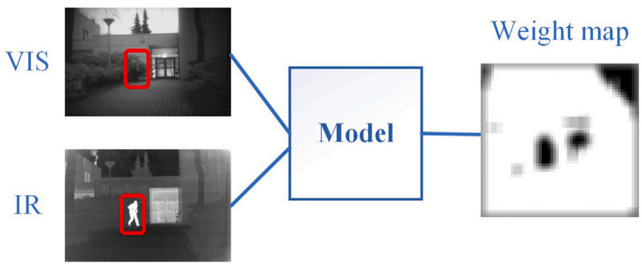

Fig. 7. The schematic of weight map generation: large weight (whiter pixel) means more contribution from visible image and small weight (darker pixel) means more contribution from infrared image.

Enhanced examples of the infrared images are shown in Fig. 9, from which it can be observed that the enhanced infrared images clearly show equal illumination.

3.5. Fusion scheme

The fusion scheme was simple. We only used the weighted average of both the enhanced visible image I_v^e and the infrared image I_i^e to obtain a fused image I_{fused} with weights from weight map W generated in Section 3.3 as follows:

$$I_{fused} = W \cdot I_v^e + (1 - W) \cdot I_i^e \tag{7}$$

Weight map W indicates the contribution of the input images to the final fused image. A large weight indicates a greater contribution from the visible image because a large weight pixel means that the pixel in the visible image is clearer than that in the infrared image. However, a smaller weight indicates a greater contribution from infrared image.

3.6. Enhancement of fused image

We used SSR [48] to further enhance the fused image I_{fused} as:

$$I_{fused}^e = \log I_{fused} - \log \left(F \circledast I_{fused} \right), \tag{8}$$

where I_{fused}^e, $\circledast$, and F are the enhanced fusion image, convolution operator and Gaussian filter, respectively. The Gaussian filter F is defined as

$$F(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(\frac{-\left(x^2 + y^2\right)}{\sigma^2} \right), \tag{9}$$

where (x, y) are pixel coordinates in the integer and σ^2 is set to 16 in this study.

4. Experimental results

4.1. Experimental setup

To demonstrate the performance of our proposed method, we used three public IVF benchmark datasets: the TNO [49], LLVIP [50], and VOT2020-RGBT [51]. They contained 21, 100, and 40 pairs of infrared and visible images, respectively.

The proposed method was evaluated both qualitatively and quantitatively. because one metric is not sufficient to measure the image quality, we choose 11 metrics and other 17 typical methods for comparison. The other 17 methods are cross bilateral filter (CBF) [52], sparse convolutional representation (ConvSR) [53], Structaware [54], DCHWT [5], HybridMSD [55], DenseFuse [56], NestFuse [57], DDcGan [58], MEF-GAN [59], Image fusion network based on proportional maintenance of gradient and intensity (PMGI) [60], CNN [46], MDlattLRR [61], multilayer deep features fusion method (VggML) [62], image fusion framework based on convolutional neural network (IFCNN) (elementwise-maximum) [63], unified unsupervised image fusion(U2Fusion) [64], ResNet-Zca [65], and residual fusion network for infrared and visible images(RFN-Nest) [66]. The comparative results from these 17 methods are obtained by running their open-source codes with recommended setting up in their corresponding studies.

because the pipeline proposed in this study is a mixed method that combines modern deep learning with traditional approaches, the typical methods chosen for comparison also cover modern and traditional methods: (1) classical image processing-based techniques that do not rely on deep learning, such as CBF [52], DCHWT [5], and HybridMSD [55]. They used filter or image decomposition techniques to extract and fuse the image features. (2) methods based on an autoencoder for deep learning, such as DenseFuse [56], NestFuse [57], and RFN-Nest [66]. They employed an encoder for feature extraction and feature fusion at the latent layer. (3) Methods based on GAN for deep learning, such as DDcGAN [58] and MEF-GAN [59]. The network consists of a generator and discriminator. The generator generates the fused image and the discriminator pushes the generator to generate a better fused image. (4) Methods based on deep learning with a one-step fusion network, such as CNN [46], VGGML [62], PMGI [60], and U2Fusion [64].

The chosen 11 metrics are edge intensity (EI) [67], cross entropy (CE) [68], spatial frequency (SF) [69], entropy (EN) [70], quality of images ($Q^{ab/f}$) [67], spatial frequency sum of correlation coefficients (SCD) [71], noise in images ($N^{ab/f}$) [68], mutual information between images ($M^{Iab/f}$) [72], standard deviation of image (SD) [73], definition (DF), and a novel Visual information fidelity (VIFF) [74].

A larger EI and SF indicate more edges and detailed information in the fused images. A larger EN value indicates more information in the fused image. Larger SCD and MI indicate that the fused image retains sufficient information from the source images and is more similar to the source images. A larger VIFF value indicates that the fused image is more natural or realistic. The large DF and SD values indicate that the contrast of the fused images was high. A large $Q^{ab/f}$ value indicates good image quality. Large $N^{ab/f}$ means a noisy image. A large CE value indicates a significant difference between the source and fused images. In summary, large metrics values, except for CE and $N^{ab/f}$, mean better-fused images.

We used Linux Ubuntu OS, Intel(R) Xeon(R) CPU E5-2687 W v3 @ 3.10 GHz, and Nvidia GeForce RTX 2080 TI GPU for training and testing. The algorithm was implemented based on the PyTorch framework. The network was trained using stochastic gradient descent (SGD) with a learning rate of 0.0002, the momentum of 0.9, and the weight decay of 0.0005. We used a step learning rate schedule with a step size of 1 and a gamma value of 0.9. The cross-entropy loss was used as the loss criterion and batch normalization was used to train the model. The training took approximately 12 h with 30 epochs. The training accuracy was 99.45% for the 1,000,000-training patches and the validation accuracy was 99.36.% for the 300,000 validation patches.

4.2. Ablation study

To demonstrate effectiveness of the proposed image enhancement components in our pipeline, four ablation pipeline experiments were performed: (1) no enhancement; (2) enhancing only the input image of the weighted averaging operation; (3) enhancing input images of both trained CNN and weighted averaging operation; (4) enhancing input images of both trained CNN and weighted averaging operation and fused image (proposed method in this study). These four ablation studies were denoted as “Abl. 1”, “Abl. 2”, “Abl. 3” and “Abl. 4”, which are shown in Figs. 10(a), (b), (c), and (d), respectively.

We comparatively show some example results from these four ablation experiments and three datasets in Figs. 11 12 13. This can be observed from Figs. 11 12 13 that interesting objects (such as persons,

Fig. 8. Some examples of enhanced visible images. First row: original visible images; second row: enhanced images.

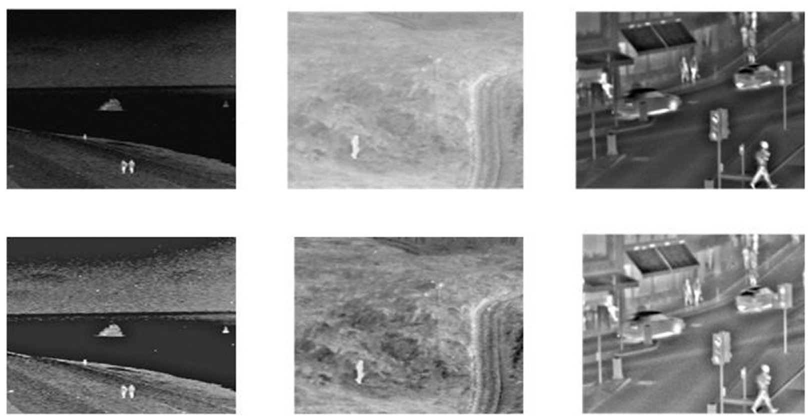

Fig. 9. Some examples of enhanced infrared images. First row: original infrared images; second row: enhanced infrared images.

Table 1
Quantitative comparative results in TNO datasets: The best, second-best, and third-best values in the tables are highlighted in red bold, green italics, and underlined blue, respectively. ↑: larger is beter; ↓: smaller is beter.

Methods	EI↑	CE↓	SF↑	EN↑	Qab/f ↑	SCD ↑	Nab/f↓	MIab/f↑	SD↑	DF↑	VIFF↑
CBF [52]	52.30	1.32	13.09	6.83	0.45	1.32	0.06	13.66	34.65	6.46	0.28
ConvSR [53]	46.02	**0.82**	16.16	7.07	0.50	1.04	0.04	14. 15	45.18	5.62	0.29
StructAware [54]	44.91	0.92	11.20	7.05	**0.62**	1.26	*0.01*	14.11	40.31	5.41	0.35
DCHWT [5]	35.76	1.43	9.55	6.53	0.40	1.52	0.02	13.07	30.00	4.67	0.28
HybridMSD [55]	46.32	1.32	12.35	6.71	0.50	1.58	0.04	13.42	31.27	6.10	0.44
DenseFuse [56]	36.48	1.40	9.32	6.85	0.47	1.53	0.04	13.71	38.04	4.62	0.39
NestFuse [57]	38.44	1.45	9.71	6.89	0.50	1.58	0.05	13.77	38.33	4.91	0.37
DDcGAN [58]	54.82	1.10	12.84	*7.42*	0.36	1.38	0.13	*14.84*	46.73	6.94	0.46
MEF-GAN [59]	36.91	1.50	7.85	6.97	0.21	1.31	0.08	13.95	43.73	3.87	0.32
PMGI [60]	37.21	1.57	8.72	6.87	0.38	1.57	0.03	13.74	33.02	4.43	0.42
CNN [46]	44.83	*0.89*	11.65	7.06	*0.58*	*1.60*	0.02	14. 13	44.92	5.7266	0.4504
MDLatLRR [61]	*86.89*	1.02	*21.62*	7.09	0.32	1.56	0.30	14.18	43.00	*11.07*	*0.88*
VGGML [62]	24.01	1.53	6.41	6.18	0.33	1.60	**0.00**	12.37	22.71	3.15	0.25
IFCNN [63]	41.56	1.60	10.85	6.59	0.51	**1.62**	0.03	13.18	31.33	5.27	0.32
U2Fusion [64]	48.49	1.33	11.04	6.72	0.39	1.59	0.08	13.45	31.38	5.83	0.53
ResNet-ZCA [65]	26.24	1.62	5.92	6.65	0.10	0.19	0.06	13.31	*46.93*	2.59	0.04
RFN-Nest [66]	29.15	1.67	6. 13	6.84	0.34	1.50	0.02	13.68	35.27	3.07	0.48
Ours	**92.80**	1.23	**27.72**	**7.89**	0.25	1.40	0.32	**15.48**	**68.75**	*10.73*	**1.22**

buildings, and cars) are more focused on in the fourth ablation study (our pipeline in this study) in the last row than in other studies and that the fused results of the fourth ablation (our method) experiment are more pleasing than the others.

We also show the quantitative comparative results of these four ablation studies in Table 2. One of the four ablation methods (the method used in this study) was better than the other methods.

Based on the qualitative and quantitative comparison of the four ablation experiments, the image enhancement components play a very important role in the fusion process, and our proposed pipeline, consisting of three image enhancement components (in ablation 4), performs the best among the four ablation studies.

4.3. Qualitative results

4.3.1. Visual example results across three datasets

First, we show our qualitative results for some example images from the TNO, LLVIP, and VOT2020-RGBT datasets in Figs. 14, 15,

Table 2
Quantitative comparison among four ablation studies: Abl. 1, Abl. 2, Abl. 3, Abl. 4. The best results are highlighted in red and bold, and the second best in green and italic. ↑: larger is beter; ↓: smaller is beter.

Ablation	Dataset	EI↑	CE↓	SF↑	EN↑	Qab/f ↑	SCD ↑	Nab/f↓	MIab/f↑	SD↑	DF↑	VIFF↑
Ab. 1	TNO	32.38	*1.69*	8.06	6.81	**0.38**	*1.58*	**0.02**	13.61	*34.18*	3.63	0.44
	VOT	34.29	*1.76*	8.97	6.91	**0.40**	*1.56*	**0.02**	13.82	*36.96*	3.89	0.45
	LLVIP	47.19	**0.53**	15.46	7.34	**0.56**	0.84	**0.01**	14.68	43.63	5.31	0.41
Abl. 2	TNO	36.50	2.21	8.87	6.76	0.33	**1.60**	*0.06*	13.51	30.38	4.13	*0.47*
	VOT	36.67	2.48	9.30	6.83	0.35	**1.56**	*0.04*	13.66	32.61	4.23	*0.47*
	LLVIP	51.31	1.52	16.28	7.37	0.40	*1.25*	*0.02*	14.74	43.63	7.14	*0.45*
Abl. 3	TNO	*38.61*	2.00	*9.46*	*6.82*	*0.34*	1.58	0.07	*13.65*	31.78	*4.40*	0.47
	VOT	*38.97*	2.30	*9.80*	*6.91*	*0.36*	1.55	0.05	*13.82*	34.08	*4.54*	0.48
	LLVIP	*54.13*	1.70	*17.40*	*7.38*	*0.41*	1.23	0.02	*14.76*	44.08	*7.67*	0.44
Abl. 4	TNO	**92.80**	**1.23**	**21.21**	**7.74**	0.25	1.40	0.32	**15.48**	**68.75**	**10.73**	**1.22**
	VOT	**89.38**	**1.22**	**21.46**	**7.70**	0.26	1.44	0.28	**15.41**	**69.67**	**10.57**	**1.26**
	LLVIP	**87.59**	*0.98*	**27.72**	**7.89**	0.37	**1.52**	0.10	**15.78**	**69.19**	**12.19**	**0.70**

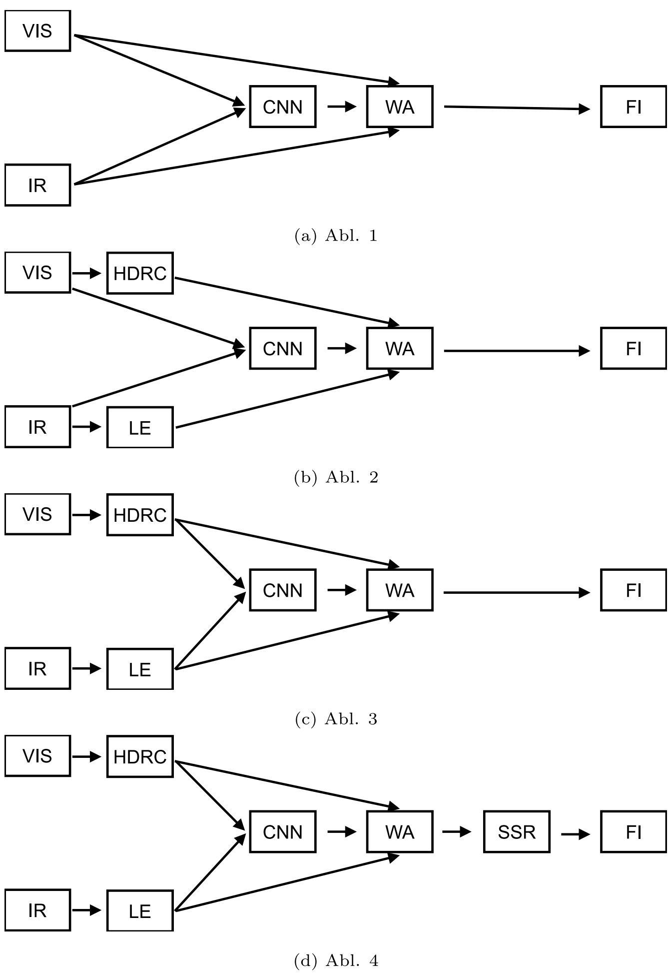

Fig. 10. Four ablation pipeline experiments. VIS: visible input image; HDRC: high dynamic range compression enhancement; IR: infrared input image; LE: linear enhancement; CNN: convolutional neural network; WA: weighted average; SSR: single scale retinex enhancement; FI: fused image. (a) No Enhancement; (b) Only enhancing input image of weighted averaging operation; (c) Enhancing input images of both trained CNN and weighted Averaging Operation; (d) Enhancing input images of both trained CNN and weighted averaging operation, and also enhancing fused image (proposed method in this study).

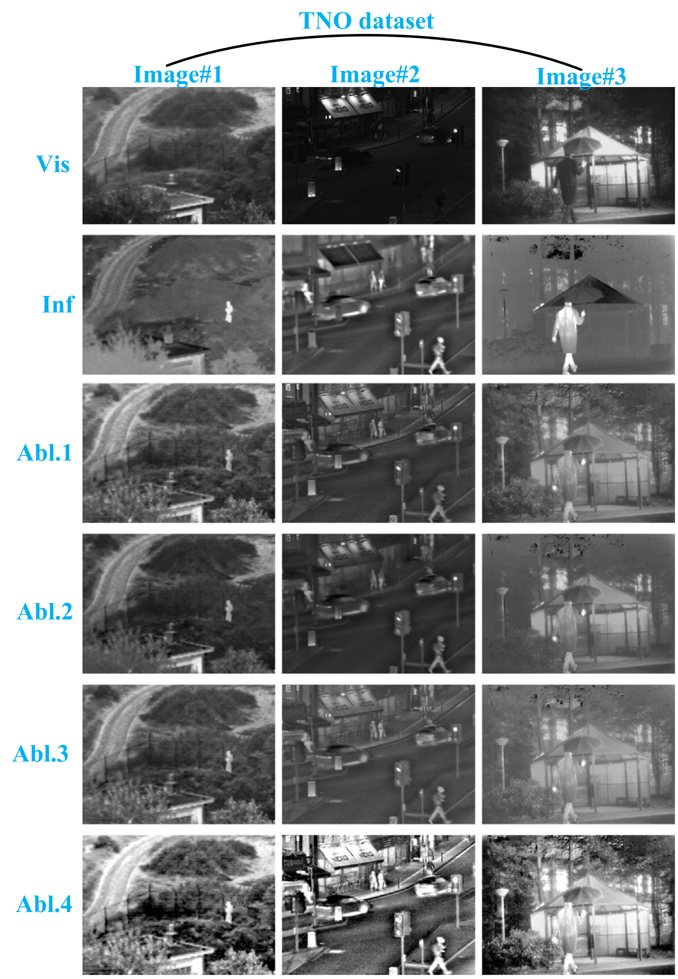

Fig. 11. Example results from four ablation experiments in TNO dataset. From top row to bottom row: visible images(Vis), infrared images(Inf), first ablation(Abl.1), second ablation(Abl.2), third ablation(Abl.3), and fourth ablation(Abl.4).

16, 17 and 18, 19 respectively. In each figure, visible input images, infrared input images, enhanced infrared images, enhanced visible images, generated weight maps, fused images, and enhanced fusion images are shown from top to bottom row, respectively.

Figs. 14, 15 show that our fused images have a much better appearance than the original input images in the top two rows. Furthermore, salient objects from both the infrared and visible images were retained in our fused images. For example, persons in Images 1, 3, 4, 6, and 7 were detected in the weight map (dark parts) and preserved in the final fused images. The lake and boat in Images 2 and 5 were also preserved in the final fused images. Based on the weight maps in the fifth row,

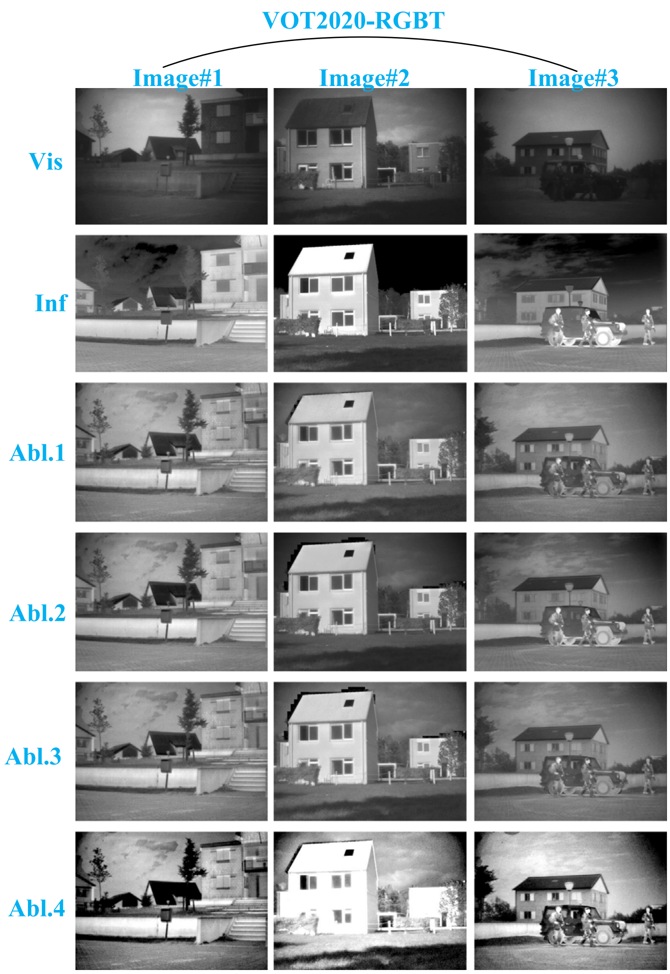

Fig. 12. Example results from four ablation experiments in VOT2020-RGBT dataset. From top row to bottom row: visible images(Vis), infrared images(Inf), first ablation(Abl.1), second ablation(Abl.2), third ablation(Abl.3), and fourth ablation(Abl.4).

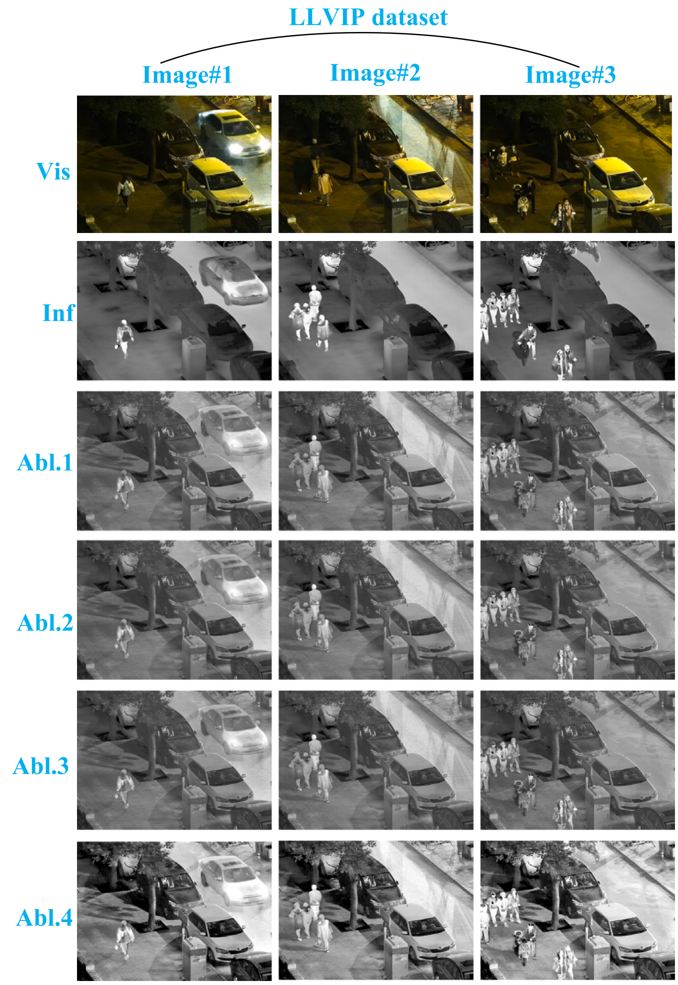

Fig. 13. Example results from four ablation experiments in LLVIP dataset. From top row to bottom row: visible images(Vis), infrared images(Inf), first ablation(Abl.1), second ablation(Abl.2), third ablation(Abl.3), and fourth ablation(Abl.4).

it can be observed that the proposed deep blur detection network can accurately detect salient objects in infrared images (dark parts in the weight maps).

From Figs. 16, 17, it can be observed that the lighting is poor, and the human bodies are more apparent in the infrared images than in the visible images, and our fused images can preserve both human body information from the infrared images and background information from the visible images. From the weight maps in the fifth row, it can be observed that the proposed deep blur detection network can accurately detect salient objects in infrared images (dark parts in the weight maps). The weight map in the last column indicates that people riding motorcycles were missed in the infrared image. This is because the people riding motorcycles are also clearly visible. Our method preserves motorcycle's riding motorcycles from visible images rather than from infrared images.

From Figs. 18, 19 we observe, our fused images retain salient objects from both infrared and visible images and our fused images and appear much better than the original images. Based on the weight maps in the fifth row, our proposed deep blur-detection network can accurately detect salient objects in infrared images (dark parts in the weight maps).

From the visual results for the three datasets, it can be observed that our method can achieve consistent results across multiple datasets. The salient objects in both infrared and visible images can be accurately preserved

4.3.2. *Visual exemplar comparison results*

To visually compare our method with other state-of-the-art methods, we present the comparative visual example results of our method with other 17 methods in Figs. 20, 21, and 22.

In Fig. 20 we observe, all these methods can preserve the salient "man" (shown in the yellow box) from the infrared image in the final fused images. However, our method can accurately preserve the background and shape of the "man". Our method does not introduce additional artifacts into the "man" or the background of the "man or its background". In contrast, some artifacts in "man" can be seen in other methods, such as in sub-figures (C), (D), (J), and (N). The artifacts in the background of the "man" are apparent from other methods, such as in sub-figures (C), (D), (E), and (F). Overall, the fused image was better than the combined results from other methods in terms of appearance and detailed information. For example, the details of our fused image in the red box ("street way") and green box ("plants") are clearer than those of the other methods. It is apparent that many artifacts were introduced in the other parts of the other methods, such as in the sky, in the tree behind the building, and in the door. Although, our image enhancement steps can also be used to enhance the fused images from other methods to obtain a better appearance (e.g., better

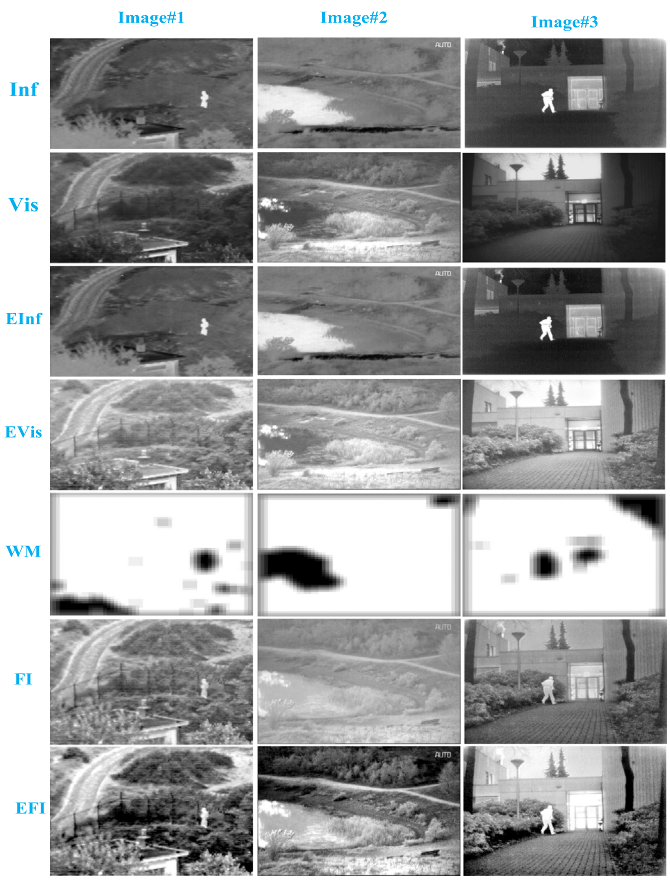

Fig. 14. Example results in TNO dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

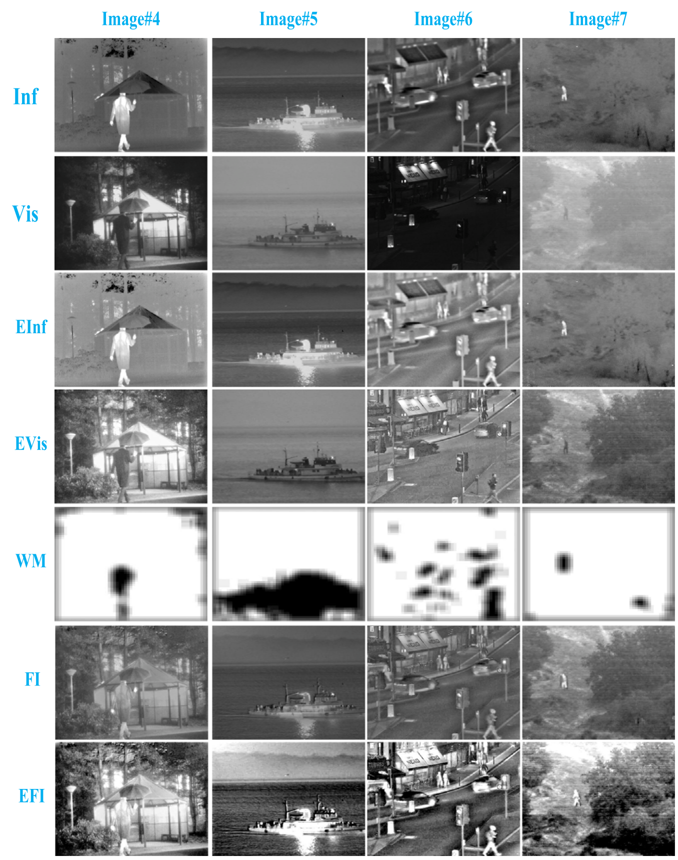

Fig. 15. Example results in TNO dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

contrast). The lost details in the fused images from the other methods could not be recovered using these enhancement steps. Therefore, our proposed IVF pipeline achieves better fused images because our proposed fusion method can preserve true details from input images without introducing extra artifacts, and the proposed enhancement steps in our pipeline can improve the overall appearance of the fused images.

According to Fig. 21, our fused image is better than the fused results from the other methods in terms of appearance and detailed information. The tree in the green box deteriorates in the fused results from all other methods and is accurately preserved in our method. The "garden lamp" in the yellow box was preserved in all methods. The "plants" in the red ellipse have apparent artifacts in the results from other methods, such as in sub-figures (C), (D), (E), (G), (H), (I), (L), (N), and (P). The proposed method can accurately preserve the "plants" in the red ellipse. The "umbrella" term can be reserved in all methods. However, additional artifacts in the "umbrella man" are evident in sub-figures (C), (D), and (F). Therefore, our proposed IVF pipeline has a strong ability to preserve details and improve the overall appearance of the fused images.

From Fig. 22, it can still be observed that our fused image is better than the fused results from the other methods in terms of appearance and detail information. The "person with backpack" in the yellow box is preserved in all methods, including ours. The "table and stand chairs" in the red box can be seen in our fused image and are difficult to see in the fused images from all other methods. The "house" in the green box is well preserved in the fused image. However, the details of the "house" are lost in other fused images from the other methods. Some artifacts were introduced to the "house" in the green boxes in other fused images from the other methods. For example, the windows at the top left of the green box are clear in our fused image and are not clear in the fused images from other methods. Our fused image is more pleasant to the human eyes than other fused images.

In summary, these visual comparative results illustrate good performance of our proposed IVF pipeline in preserving true input details without introducing extra artifacts.

4.4. Quantitative results

To quantitatively compare the fusion performance of the proposed IVF pipeline with the other 11 algorithms, comparative metrics values from different methods are listed in Table 1 in the TNO dataset. The best, second-best, and third-best values in the tables are highlighted in red bold, green italics, and underlined blue, respectively.

As shown in Table 1, the proposed method achieved the six best metrics (EI, SF, EN, MI, SD, and VIFF) and one second-best metrics (DF) among all 17 methods, including ours. The largest EI, SF, EN, and MI values indicate that our method extracts most of the edge information and detailed information from the source images. The largest VIFF value indicated that our fused image was the most realistic and natural. The relatively large DF and SD values indicate a high contrast in the fused image.

We also show the standard deviations of all the metrics over the TNO dataset in Table 3. Notably, it is inappropriate to use these standard deviations to justify the performance of methods that take

Table 3
Standard deviation in TNO datasets. Smallest, second smallest, and third smallest are red bold, green italics, and underlined blue, respectively.

Methods	EI	CE	SF	EN	Qab/f	SCD	Nab/f	Mlab/f	SD	DF	VIFF
CBF [52]	13.84	0.88	3.96	0.48	0.08	0.29	0.05	0.97	11.84	1.96	0.12
ConvSR [53]	16.13	*0.64*	4.77	0.46	0.08	0.30	0.02	0.92	12.66	2.18	0.13
StructAware [54]	16.51	0.78	4.21	0.43	0.06	0.35	*0.01*	0.87	10.65	2.35	0.17
DCHWT [5]	14.07	0.80	4.27	0.55	0.08	0.25	*0.01*	1.10	10.24	2.19	*0.09*
HybridMSD [55]	17.54	1.05	4.31	0.52	0.08	*0.19*	*0.01*	1.04	12.52	2.46	0.10
DenseFuse [56]	*10.86*	0.83	3.01	0.34	0.07	0.20	0.06	0.68	11.18	1.73	0.18
NestFuse [57]	12.90	0.94	3.47	0.34	0.07	*0.19*	0.08	0.68	11.63	1.89	0.15
DDcGAN [58]	17.89	0.74	4.40	**0.12**	0.09	0.23	0.17	**0.24**	*4.91*	2.63	0.27
MEF-GAN [59]	**10.01**	0.96	**1.74**	0.55	*0.05*	0.41	0.05	1.11	13.81	**1.06**	0.17
PMGI [60]	14.23	0.94	3.22	0.29	*0.05*	0.26	0.04	0.58	6.51	2.08	0.19
CNN [46]	16.35	0.81	4.12	0.45	0.09	0.22	*0.01*	0.90	13.06	2.32	0.11
MDLatLRR [61]	26.72	1.10	6.95	0.55	0.09	0.39	0.05	1.10	10.11	3.85	0.19
VGGML [62]	16.23	0.91	4.02	0.50	0.07	0.22	0.02	0.99	9.80	2.35	*0.09*
IFCNN [63]	18.79	0.95	3.89	0.41	0.06	0.22	0.07	0.82	7.78	2.56	0.20
U2Fusion [64]	13.63	1.02	*2.61*	0.43	0.10	0.30	0.02	0.87	10.08	*1.56*	0.19
ResNet-ZCA [65]	17.06	0.86	4.33	0.41	**0.01**	**0.16**	**0.00**	1.63	12.84	6.09	**0.04**
RFN-Nest [66]	16.51	0.83	4.24	0.43	0.07	0.36	*0.01*	0.86	10.31	2.36	0.12
Ours	24.42	**0.53**	5.40	*0.19*	0.08	*0.19*	0.16	*0.38*	**4.38**	3.12	0.76

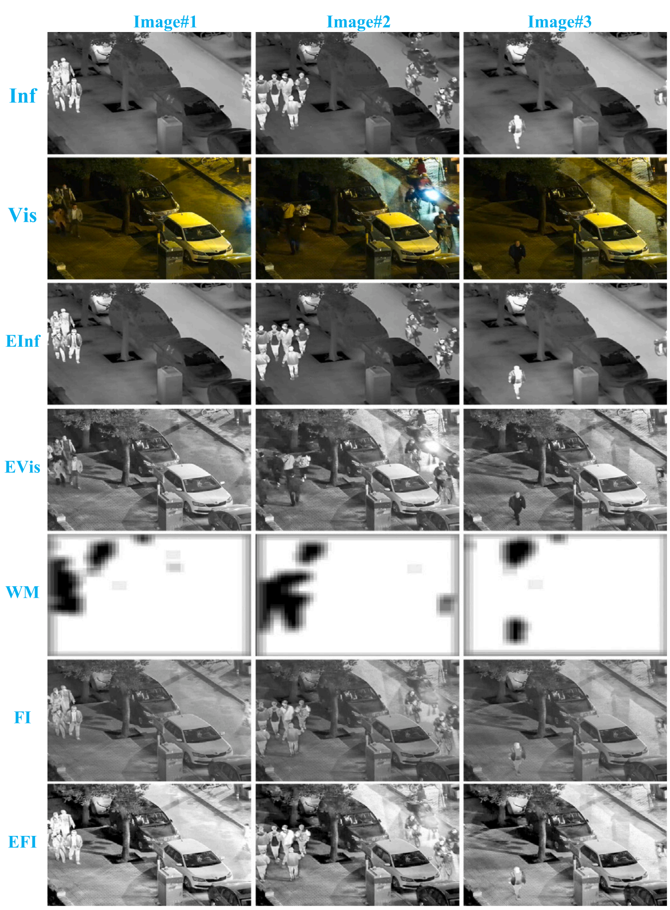

Fig. 16. Example results in LLVIP dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

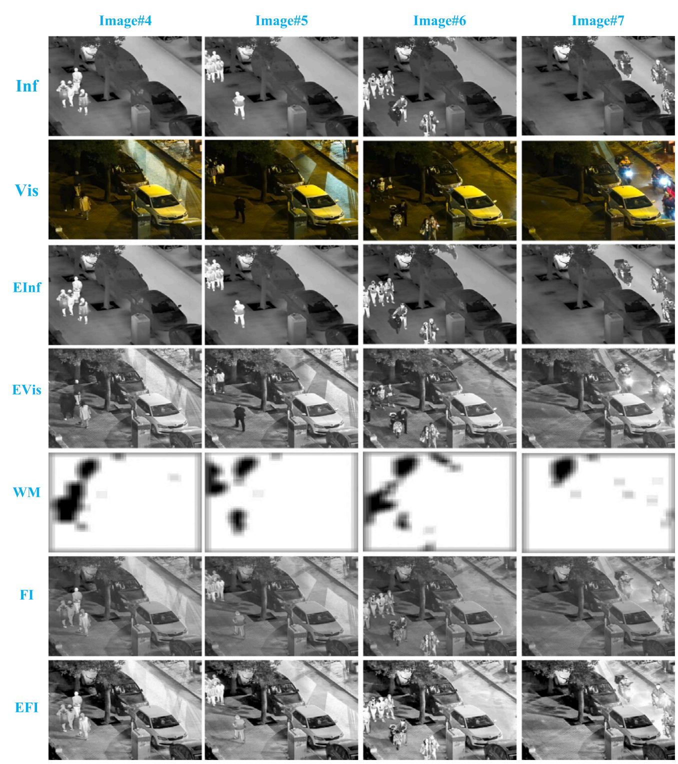

Fig. 17. Example results in LLVIP dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

significantly different metric measurement ranges. However, the standard deviation may indicate the robustness of methods that use similar metric measurement ranges.

For example, although our proposed method has a relatively large "EI" standard deviation "24.42" (shown in the entries in the last row and second column in Table 3) than the "EI" standard deviation "10.01" from "MEF-GAN" method [59] (shown in the entry in the tenth row and second column of Table 3), which does not imply that our proposed method is more robust than the "MEF-GAN" method [59]. Because the proposed method has a much larger "EI" value of 92.8047 (shown in the entry in bottom row and second column in Table 1) than the "EI" value "36.905" taken by "MEF-GAN" method [59] (shown in the entries in the tenth row and second column in Table 1). However, although the "MDLatLRR" method [61] has smaller "EI" value "86.8855"(shown in the entry in the thirteenth row and the second column in Table 1) than our method's "EI" value "92.8047" (shown in the entry in the last row and the second column in Table 1), it has larger "EI" standard deviation "26.72" (shown in the entry in the thirteenth row and the second column in Table 3) than our "EI"

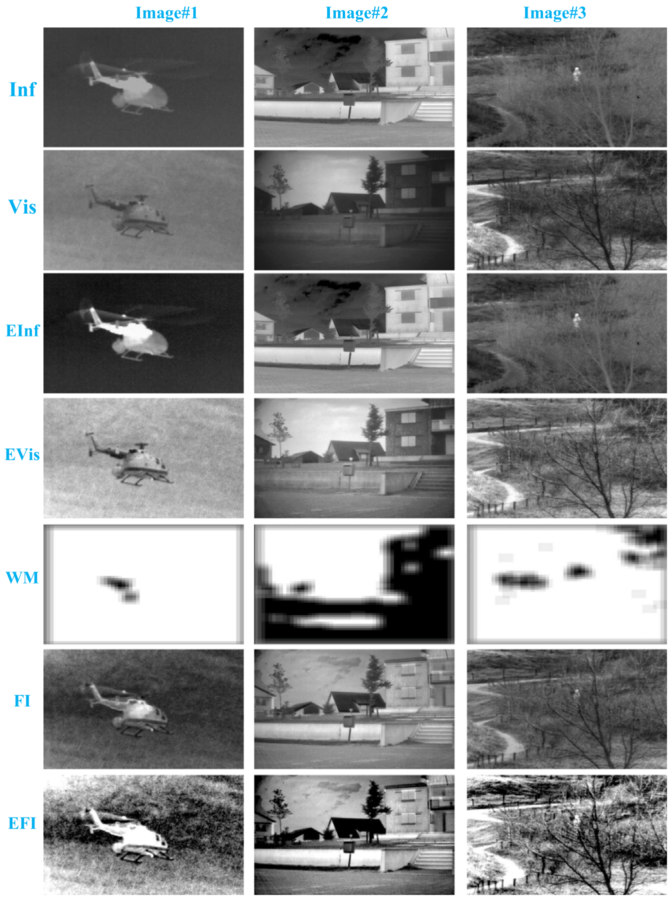

Fig. 18. Example results in VOT2020-RGBT dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

Fig. 19. Example results in VOT2020-RGBT dataset: from top row to bottom row: input infrared images(Inf), input visible images(Vis), enhanced infrared images(EInf), enhanced visible images(EVis), weight maps(WM), fused images(FI), and enhanced fusion images(EFI).

standard deviation "24.42" (shown in the entry in the last row and the second column in Table 3). In other words, our method is more robust than the "MEF-GAN" method [59]. Similarly, it is not difficult to roughly compare the robustness of other methods and our proposed method based on the mean value Table 1 and the standard deviation Table 3.

In summary, the pipeline proposed in this study achieved superior performance in both quantitative and qualitative results when compared with other state-of-the-art methods.

5. Conclusion

In this study, we proposed a simple and fully automatic IVF pipeline (no human intervention) that combines modern deep learning with traditional fusion methods and image enhancement. This pipeline was easy to understand and interpret. We converted the normal unsupervised IVF problem into a supervised classification problem and verified the assumption that a modality with a sharp image provides more information than a modality with a blurred image. It was demonstrated that our proposed cross-domain training strategy works well in the IVF domain and that infrared images are not required in the training phase. To the best of our knowledge, this is the first study that combines supervised deep binary classification with image enhancement in the IVF domain. Extensive experimental results demonstrate the superior performance of our proposed IVF pipeline to other state-of-the-art methods. However, the method proposed in this study processes the input image patch by patch. Extending our method to work on an entire image rather than on a patch (similar to fully convolutional semantic segmentation) is a potential future research direction.

CRediT authorship contribution statement

Jin Qi: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Deboch Eyob Abera:** Data curation, Software, Writing – original draft. **Mola Natnael Fanose:** Investigation, Visualization. **Lingfeng Wang:** Software, Validation. **Jian Cheng:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data and codes are publicly available.

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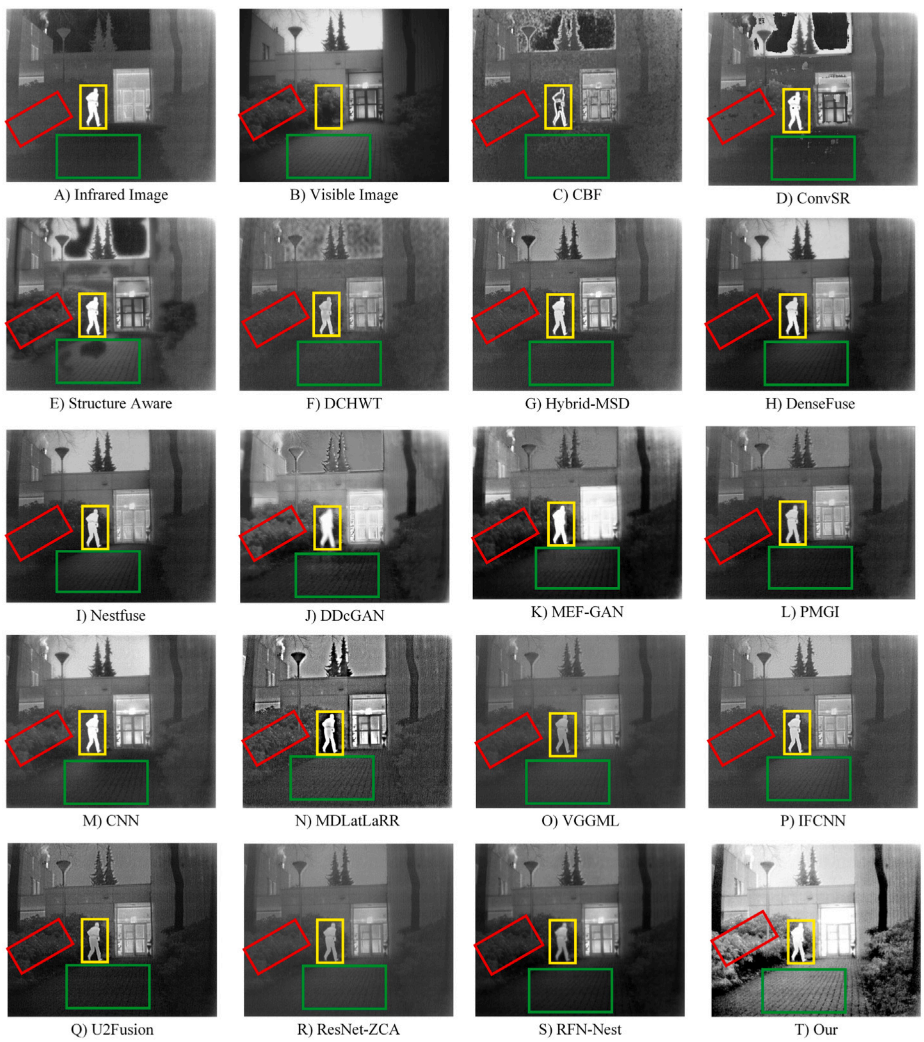

Fig. 20. Comparative results in the “man” image. Red box: street way; yellow box: man; green box: plants.

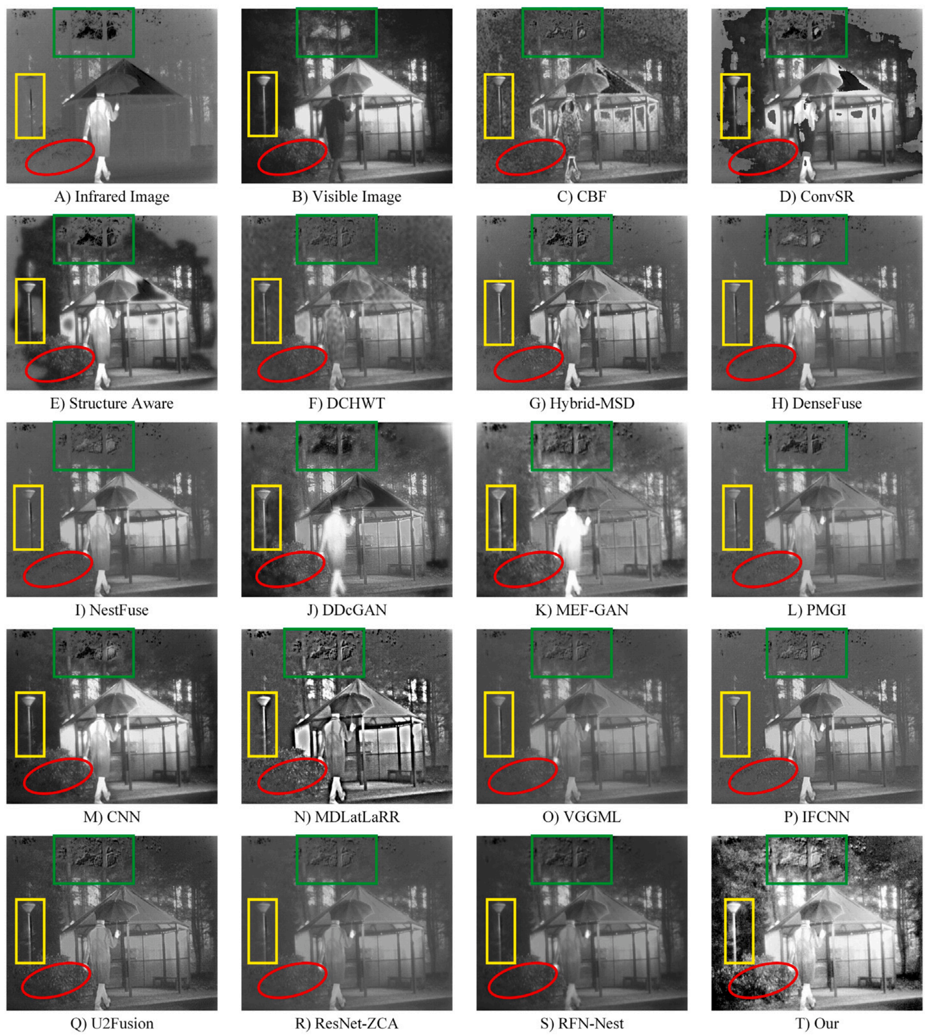

Fig. 21. Comparative results in the "umbrella man" image. Red ellipse: plants; yellow box: garden lamp; green box: tree.

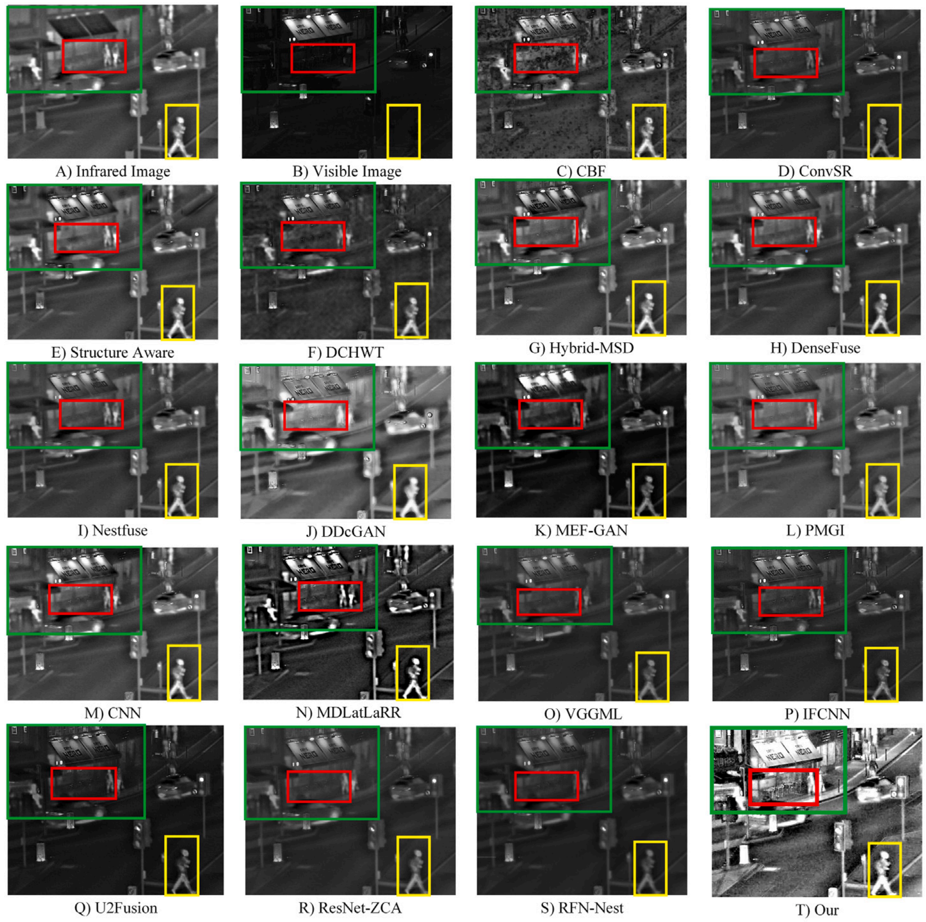

Fig. 22. Comparative results in the "man on the street" image. Red box: table and stand chairs; yellow box: person with backpack; green box: house.

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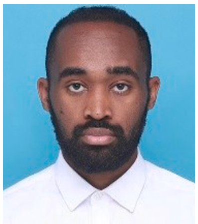

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